

Srivani Balakrishna

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Data Analyst

PROFESSIONAL SUMMARY:

- Over 4 years of experience in analyzing, interpreting, and visualizing large datasets for finance, healthcare, and enterprise systems using Python, SQL, and BI tools like Power BI, Tableau, and Looker.
- Proficient in cleaning, transforming, and enriching high-volume data using pandas, Power Query, and Excel; delivered actionable insights through automated processes, improving reporting speed and accuracy across departments.
- Expertise in SQL Server and PostgreSQL for writing optimized queries, stored procedures, and functions; skilled in data modeling techniques including star schema and normalization for decision support systems.
- Built and maintained end-to-end ETL pipelines using Azure Data Factory, SSIS, and Python scripts to support dynamic reporting needs and ensure timely, accurate, and governed data flows.
- Integrated Google BigQuery with Cloud Functions and Vertex AI to scale advanced analytics, machine learning model deployment, and predictive forecasting across real-time customer and operational datasets.
- Designed and deployed event-driven streaming data pipelines using GCP Dataflow and Pub/Sub, supporting real-time insights and anomaly detection across finance and operations verticals.
- Engineered cloud-native data lakes using Google Cloud Storage and BigQuery for cost-effective storage and analytics, enabling fast processing of structured and semi-structured enterprise data.
- Developed Looker dashboards using complex joins, dimensions, and measures from BigQuery; created real-time, role-based reports to support marketing, product, and finance leadership teams.
- Automated reporting pipelines with Cloud Composer and App Script, integrating Google Sheets and GCP services for seamless workflows and stakeholder-accessible dashboards with built-in error handling.
- Created machine learning models in scikit-learn for churn prediction and ROI optimization; partnered with leadership to define KPIs and implement data-driven strategies across campaign lifecycles.
- Applied Agile and Scrum best practices using Jira and Git to deliver iterative enhancements in dashboards, data pipelines, and ETL frameworks; drove transparency and efficient sprint delivery.
- Conducted A/B testing and statistical hypothesis validation using Python, statsmodels, and Excel; produced detailed reports that improved product decision accuracy and marketing ROI.
- Wrote modular Python scripts for automated ingestion, cleansing, and transformation of third-party and internal data, drastically improving pipeline reliability, auditability, and development velocity.
- Optimized SQL-based monthly reporting jobs, reducing execution time by 40% and eliminating redundancy through automation, parameterization, and incremental load strategies using SQL and Power BI.

TECHNICAL SKILLS:

- **Programming** - Python, SQL, R
- **Data Visualization** - Power BI, Tableau, Looker, Excel
- **Cloud Platforms** - Google Cloud Platform (BigQuery, Dataflow, Cloud Storage, Vertex AI), Azure
- **ETL Tools** - Azure Data Factory, SSIS
- **Databases** - SQL Server, PostgreSQL, MySQL
- **Data Wrangling** - Pandas, NumPy, OpenRefine
- **Version Control** - Git
- **Others** - Jira, Confluence, REST APIs, Airflow (Cloud Composer), Excel Power Query

PROFESSIONAL EXPERIENCE:

Northern Trust Corp

Nov 2024 – Present

Data Analyst

Responsibilities:

- Designed automated Power BI dashboards for enterprise KPIs, connecting SQL Server and BigQuery to deliver real-time insights and reduce manual reporting delays, improving efficiency and transparency across business functions.
- Partnered with finance teams to build dynamic financial dashboards that monitored spend, variance, and projections, leading to improved budget accountability and a 20% rise in data-driven investment decisions.
- Developed scalable ETL pipelines using Google Cloud Composer, Python, and BigQuery SQL to ingest and transform transactional data, enabling near real-time analytics and improved performance for operational teams.
- Built anomaly detection models using scikit-learn to flag irregularities in transaction streams, reducing fraud detection false positives by 30% and improving audit response and risk mitigation.
- Migrated financial datasets from SQL Server to BigQuery using Cloud Storage, resulting in faster query performance, centralized data access, and simplified schema management for cross-functional analytics.
- Automated monthly reporting cycles using Python and Airflow, decreasing manual data handling by 15 hours per week and reducing data refresh failures through exception handling and scheduling logic.
- Collaborated with risk stakeholders to embed Basel III compliance logic into BigQuery models, supporting accurate exposure calculation and ensuring consistency in regulatory reporting pipelines.
- Refined GCP warehouse schema with engineering teams by normalizing historical data, enforcing data quality checks, and standardizing naming conventions to improve downstream compatibility and model accuracy.
- Created asset performance dashboards in Looker with scheduled data delivery, enabling portfolio managers and analysts to track return metrics, drawdowns, and risk-adjusted benchmarks daily.
- Developed variance dashboards using advanced DAX in Power BI, allowing leadership to visualize trends, conduct variance analysis, and apply filters dynamically for operational review meetings.
- Extracted marketing and engagement data from REST APIs, transforming it using Python and loading into BigQuery staging for campaign analysis and ROI performance monitoring across product lines.
- Led knowledge-sharing workshops on BigQuery performance best practices, covering partitioning, clustering, materialized views, and query optimization techniques to improve team productivity and reduce compute costs.
- Performed data validation across systems using reconciliation scripts and row counts, ensuring 99.9% accuracy in financial reports and dashboards, and preventing data quality issues during audits.
- Supported ESG analytics initiatives by modeling sustainability scores in Python, visualizing outputs in Tableau, and aligning metrics with regulatory standards for internal and investor presentations.
- Built Cloud Logging alerts and configured failover pipelines in Cloud Composer to detect and resolve ETL job failures proactively, increasing pipeline resilience and reducing downtime in critical jobs.
- Developed secured Power BI drill-through reports with row-level access for internal audit teams, facilitating compliance reviews while maintaining user-level data isolation and traceable reporting trails.

Trustmark

Mar 2023 – Oct 2024

Data Analyst

Responsibilities:

- Designed Tableau dashboards for HR and payroll metrics by integrating ADP and SQL sources, enabling centralized workforce analytics and empowering department heads with real-time visibility into team structure and attrition.
- Wrote optimized SQL queries to analyze tenure distribution and resignation trends, generating actionable insights that improved HR's workforce planning and strategic hiring decisions across business units.
- Conducted A/B testing on internal policy communications using Python statistical libraries, guiding leadership decisions based on conversion lift, open rates, and feedback response metrics for targeted employee outreach.
- Automated legacy Excel-based reports using pandas and openpyxl, eliminating manual data entry, enhancing report reliability, and saving over 10 hours of weekly analyst labor across HR and finance functions.
- Collaborated with IT to build secure GCP access layers for HR data; developed BigQuery views with row-level security to ensure data privacy for compensation, benefits, and personnel records.
- Created self-service dashboards via Google Sheets API and App Script, empowering managers with near real-time HR data access and reducing dependency on data teams for standard requests.

- Developed time-series models for overtime cost forecasting using Python and Tableau, enabling finance teams to plan labor budgets more effectively and forecast spending fluctuations with greater confidence.
- Deployed and maintained ETL pipelines in Dataform for BigQuery, centralizing transformation logic, versioning workflows, and supporting automated data refreshes with minimal intervention.
- Standardized inconsistent job title formats using Python fuzzy matching techniques, improving internal job mapping accuracy and reducing HR system classification issues by over 35%.
- Created dashboards to track DEI (Diversity, Equity & Inclusion) policy metrics and audit compliance; presented results to HR leadership and supported monthly reporting to external auditors.
- Cleaned and harmonized third-party benefits data from multiple vendors, applying transformation and validation rules before loading into the central data warehouse for unified HR insights.
- Built sentiment analysis dashboards using Python NLP tools to interpret employee feedback from surveys, allowing HR to address morale issues and identify positive trends proactively.
- Authored detailed documentation for data sources, transformation logic, and dashboards to support onboarding, knowledge transfer, and long-term maintainability for the HR analytics team.
- Conducted weekly QA checks across pipelines and data sets, identifying upstream inconsistencies and collaborating with systems teams to resolve root causes and ensure sustained data quality.

BNP Paribas

May 2021 – Dec 2022

Associate Data Analyst

Responsibilities:

- Created automated SQL queries for credit risk exposure tracking, reducing manual reporting time by 60% and supporting high-frequency reviews across operational and compliance teams.
- Built Python scripts to clean CSV exports, enrich data from external market sources, and merge datasets to streamline analysis and reduce manual data handling overhead for risk teams.
- Designed Tableau dashboards for operational risk KPIs such as SLA adherence, incident categories, and response times, enabling clearer tracking of risk controls and compliance performance metrics.
- Conducted QA validation across daily pipeline runs; introduced anomaly detection flags to capture and alert on outlier values or discrepancies before dashboards were refreshed and distributed.
- Supported onboarding of legacy risk systems by transforming flat-file exports and staging them in PostgreSQL, validating schema integrity and minimizing data loss during system migrations.
- Collaborated with stakeholders to define business rules, sketch mockups, and iterate Tableau dashboard designs to improve usability and answer ad hoc analytical needs.
- Maintained a centralized KPI glossary and documented rules for calculations, data lineage, and caveats to ensure consistency in reporting and reduce onboarding friction for new team members.
- Developed Power BI dashboards to monitor ETL job success rates and pipeline health, allowing engineering and risk teams to troubleshoot failures and improve operational resilience.
- Automated Excel workbook consolidation tasks using Python (pandas), appending data to SQL Server tables and eliminating repetitive manual copy-paste work for monthly team metrics.
- Analyzed client onboarding durations by region and team; summarized findings in management reports to highlight bottlenecks and suggest interventions for improving service delivery timelines.
- Delivered monthly risk metric reviews to team leads, incorporating stakeholder feedback to refine KPI definitions and improve the relevance and clarity of reported figures.
- Partnered with audit teams to fulfill data lineage and metadata requests, mapping key tables and transformations to support compliance reviews and risk data governance practices.

Certifications:

- Microsoft Certified: Fabric Data Engineer Associate

Educational Details:

Master's in Computer Science - University of Massachusetts Lowell

Bachelor's in Electronics & Communication Engineering – Geethanjali College of Engineering and Technology